

- 6 (a) p.d. per unit length = $V_{XY} / 100 = 2.0/100 = 0.020 \text{ V cm}^{-1}$ C1
 XP = $1.50/0.020 = 75 \text{ cm}$ A1
- (b) $V_{XY} = (5.0/6.0) (2.0 \text{ V}) = 5/3 \text{ V}$ C1
 p.d. per unit length = $V_{XY} / 100 = (5/3)/100 = 0.017 \text{ V cm}^{-1}$ C1
 XP = $1.50/0.017 = 90 \text{ cm}$ A1
- (c) terminal p.d. of cell B = $1.50 - (1.50/1.80)(0.80)$ or $[1.0/(1.0 + 0.80)](1.5)$ C1
 = 0.833 V C1
 XP = $0.833/0.020 = 41.7 \text{ cm}$ A1

Comments: Some found this question challenging. The difficulties they experienced were in calculating voltage per unit of length of wire XY and identifying the p.d. across XP.

- 7 (a) (i) (induced) e.m.f. proportional/equal to rate of change of flux (linkage) B1